

Mathematical core (see Figure 1)

Characteristic fields
(Definition 2.3)
 $\kappa, \gamma, \mu \in [0,1]$

Operator $-\Delta - b$

$\lambda_1(-\Delta - b; M)$
(Definition 3.13)

$\lambda_1(t)$

Witnesses (Section 3.7)

Spectral capacity witness
(Definition 3.36)
 $\psi_s = 1 / (1 + \exp(\lambda_1/s))$

Temporal witness
 $\psi(t) \in [0,1]$

ψ', ψ'', ψ'''

via $M(t)$

κ, γ, μ at time t

updates
 $b(\cdot, t)$

Capacity erosion
of κ, γ, μ or $M(t)$

Deferred
(Remark 3.44)

Coherence debt
(Definition 3.38)
 $\dot{D} = \eta[\psi - \psi_s]^+ - \rho D$

Base resilience R_{base}
(Definition 3.34)

$R = R_{\text{base}} / (1 + D)$
(Definition 3.40)

$\lambda_{\text{eff}} = \lambda_0 / R$
(Definition 3.40)

via κ, γ, μ

Dynamical closure
(Definition 3.42)
 $b(x,t) = \kappa\gamma - \lambda_{\text{eff}}(t)\mu$

Debt, resilience,
effective cost

Feedback loop
(Remark 3.41)

a tendency, not analysed

Architectural interpretation
(Remark 3.43)

Feedback and interpretation (not analysed)